

OSSAMA ELHAKKI

Junior Machine Learning & AI Engineer

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• Casablanca, Morocco

♥24 yrs

🌐 ossamaelhakki.com

ABOUT ME

Junior AI Engineer with a Master's in Distributed Systems & AI and a hands-on project portfolio spanning computer vision, NLP, generative AI, and AI automation. I enjoy building end-to-end intelligent systems — from model integration to secure cloud deployment — using Python, FastAPI, Vue.js, and Docker. Strong fundamentals in ML and full-stack development, and eager to learn fast within a team where I can ship real-world AI products. Open to full-time, freelance, and remote roles worldwide.

WORK EXPERIENCE

Ofoto — Fine Art Digital Studio | February 2025 – October 2025 | Casablanca, Morocco

Full-Stack & AI Engineer (Final Year Project + Extended Contract)

- **AI image-generation platform** — built a full web app for high-quality img2img creation with Stable Diffusion (Juggernaut, Dreamshaper, RealVision) via AUTOMATIC1111.
- **Modular architecture** — Vue.js 3 + Vite + TailwindCSS frontend (touch-screen ready), FastAPI + SQLAlchemy + PostgreSQL backend, and Redis for async task queues.
- **Face-identity preservation** — integrated InstantID (IP-Adapter + InsightFace) to generate personalized AI portraits from a single reference photo at live events.
- **Security & delivery** — JWT auth with role-based access (B2C, B2B organizer, Super Admin), CSRF/XSS protection, and UML-modeled delivery across 5 sprints.
- **Secure cloud deployment** — Docker Compose on Ubuntu (AWS), Cloudflare DNS + Certbot SSL, UFW, WireGuard VPN to the A1111 API, and fail2ban — 99.9% uptime, -35% latency.
- **Performance & scale** — -40% time-to-live via optimized containerization; handled 500+ concurrent requests with cron PostgreSQL backups and Nginx gzip/static caching.

FSTM Research Laboratory | February 2023 – July 2023 | Mohammedia, Morocco

Research Intern — Attendance Management System

- **Real-time face recognition** — built a computer-vision pipeline to automatically identify registered students via live database comparison.
- **Automated reporting** — generated attendance PDF reports for any date range, eliminating manual processing.

EDUCATION

ENSET Mohammedia

Master's — Distributed Systems & AI

Honour: *Bien (Good)*

FST Mohammedia

Bachelor's — Applied Mathematics

Honour: *Assez Bien (Satisfactory)*

SKILLS

Programming Python, Java, SQL, TypeScript

Deep Learning PyTorch, TensorFlow, Keras, U-Net, EfficientNet, DCGAN, Transformers, LSTM

NLP & LLMs BERT, DistilBERT, HuggingFace, Seq2Seq, TF-IDF, Sentiment Analysis, NMT, Text Generation

Backend & APIs Spring Boot, FastAPI, Next.js, Vue.js, SQLAlchemy, Redis, REST APIs, gRPC

MLOps MLflow, KServe, Prometheus + Grafana, ONNX Export, CUDA

Machine Learning Scikit-learn, XGBoost, LightGBM, CatBoost, Optuna, SHAP, Pandas, NumPy

Computer Vision YOLOv8, OpenCV, Stable Diffusion, ControlNet, InstantID, Segmentation, Face Recognition, Albumentations

LLM Apps & Automation RAG, LangChain, LangGraph, vLLM, LLM Agents, Ollama, Pinecone, n8n, Apify, WhatsApp API

Data Engineering PostgreSQL, Supabase, MongoDB, ETL Pipelines, Spark, Airflow, Kafka, Tableau

DevOps & Cloud Docker, Kubernetes, Terraform, AWS, Nginx, Git, GitHub Actions, Jenkins, CI/CD, OAuth2/Keycloak

LANGUAGES

Arabic — Native

French — B1/B2 (Intermediate)

English — B1/B2 (Intermediate)